

EXPERIMENT-1

Basic Image Handling and Processing – Image to Gray-scale

AIM

To perform basic image handling and processing operations using Python by reading an image and converting the image from its original color format into a gray-scale image.

PROCEDURE

1. Install the OpenCV library in Python using the command: `pip install opencv-python`.
2. Import the OpenCV library using `import cv2`.
3. Read the input image using the `cv2.imread()` function.
4. Check whether the image has been read successfully.
5. Convert the color image into a gray-scale image using `cv2.cvtColor()` with the `COLOR_BGR2GRAY` conversion method.
6. Display the original image and the gray-scale image using `cv2.imshow()`.
7. Wait for a key press and close all image windows using `cv2.waitKey()` and `cv2.destroyAllWindows()`.

SAMPLE INPUT

Input image file: sample.jpg

Python program:

```
import cv2

img = cv2.imread("sample.jpg")

if img is None:
    print("Image not found")
else:
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    cv2.imshow("Original Image", img)
    cv2.imshow("Gray-scale Image", gray)
```

```
cv2.waitKey(0)  
cv2.destroyAllWindows()
```

SAMPLE OUTPUT





RESULT

Thus, the given image was successfully read using Python and OpenCV, and the color image was successfully converted into a gray-scale image.